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SCH 113: LABORATORY MANAGEMENT AND CHEMICAL ANALYSIS

Nature, Scope and Importance

"Analysis" originates from Greek words: *ana* meaning up and *lyein* meaning loosen. Analysis means to loosen up or separate into parts.

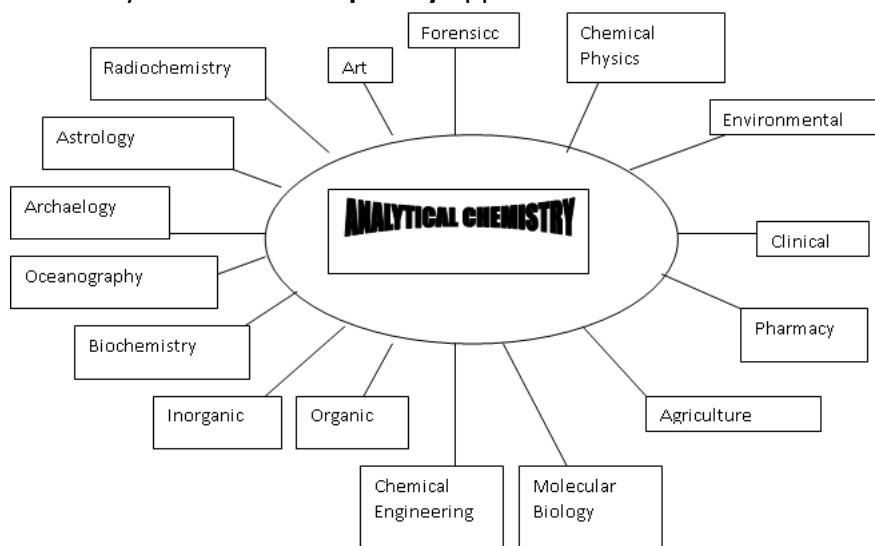
Defined:- the **identification** (qualitative measurement) and **determination** (quantitative measurement) of the relative amounts of one or more components in an **analytical sample**. It explores thus what is in a sample, and how much is in the sample.

Samples may be biological such as blood and urine or environmental such as water and soil samples.

- Name any other five examples of biological and environmental samples
.....

Position of analytical chemistry and its application

Branches of chemistry:- physical, organic, inorganic biochemical and analytical. Analytical chemistry has a **multidisciplinary** application.



Role of an analytical chemist

- i. Instrumental design
- ii. Sampling techniques
- iii. Development and evaluation of new procedures
- iv. Laboratory management
- v. Classical analysis
- vi. Statistical analysis
- vii. Optical systems development
- viii. Computer programming and electronics.

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Some examples

- **Economic value:-** determining the amount of gold in a coin or amount of protein in animal feed, the amount of vitamin or aspirin in a tablet.

- **Health hazard:** - determining the amount of CO, SO₂ in emission from charcoal burning, amount of lead from a car exhaust.
- **Disease diagnosis:** - determining glucose in urine for diabetics.
- **Relating properties to composition and structure:** - both physical and chemical properties of lubricants, plastics, pesticides, herbicides, drugs etc.
- **Research activities:** - determining accumulation and fate of pesticide in the food chain or metabolic pathways of drugs in the human body.

Steps of Analysis

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- i. Sampling
- ii. Sample pre-treatment
- iii. Conversion to desired constituents to measurable form
- iv. Measurement of desired constituent
- v. Calculation and interpretation of data. In this lesson we shall discuss these steps.

Sampling

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Sampling is done to obtain a **sample** (a portion of the substance to be analyzed) whose composition both qualitatively and quantitatively is representative of the substance being sampled. However, before starting the sampling procedure it is important to get information on the sample and the method to be used in analysis.

Factors to consider in sampling

Literature search on the sample, methods of analysis, frequency and duration of sampling (For example sampling air for analysis of lead pollution this can be done during the rush hour and between the rush hours), sampling devices (varies for solid, gaseous, liquid samples), sample size and the storage containers (shouldn't react with the sample).

Methods of sampling

These depends on the mass of sample required, nature and size of batch sampled.

Random sampling :- A sample is taken either by selecting population members at regular intervals or by use of random number table where each member of the substance is allocated a number and the numbers are picked randomly. This is done to avoid human bias. Its used for materials that are discrete, such as tablets in pharmaceutical industry.

Statistical sampling :- It follows a statistically based plan for materials that are not discrete but that are bulk and is done in two steps

- Collection of a gross sample, with its size varying
- Reduction of the gross sample to a size convenient for laboratory work through blending, grinding and many other techniques.

Sample pre-treatment

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Analytical samples are carefully handled to minimize contamination (or alteration), loss and decomposition or matrix change from atmosphere exposure during storage. For example, alkaline samples react with CO₂ in the air when not well protected; for metal analysis in water

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samples these metal may be absorbed into the walls of the container if not acidified immediately. Other pretreatments include refrigeration and drying.

Methods of converting analytes to measurable form

There are two methods:-

- **Totally destroying the sample matrix:-** For inorganic analytes or those which can be converted to an inorganic derivative for measurement and it includes acid treatment/digestion.
- **Non-destructive or only partially destructive:-** For organic analytes and done by adding appropriate reagents to the sample and then with through grinding, mixing and refluxing desired constituents of solid materials are extracted.

Measuring the Analyte

The methods used in estimation or determinations are classified depending upon:-

- a) The instrument/equipment
 - a. classical (chemical or wet)
 - b. modern (instrumental) methods.
- b) The sample size
 - a. Macro
 - b. micro analysis.

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Classical (chemical or wet) methods:- basic equipments are used based on simple chemical analysis. Eg volumetric and gravimetric analysis. The methods usually require samples with high concentrations.

Modern (instrumental) methods:- based on measurement of physical or chemical property of a substance and use of electronic, thermal or optical principles for determination. These methods can be used to analyze samples with low concentrations. Eg AAS, GLC, HPLC etc.

Comparison of classical and instrumental methods

Classical	Instrumental
Accounts for about 10% of all the current analytical work WHY? • Less sensitive (upto microgram units) • Determines one analyte at a time Small concentration range of 2-3 orders of magnitude i.e. powers of 10 • Labor intensive • Less efficient in data handling for example in recording hence, less reliable data.	Accounts for 90% of the current analytical work WHY? • Very sensitive May determine several samples compounds (multi-analyzer, even>10) More than 6 orders of magnitude i.e. power of 10 • Quicker and cheaper for large samples

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Advantages Simple procedures Procedures are often accurate Method based on absolute measurements Specialized training not required Cheaper equipment Disadvantages Storage of large volume of solution Instability of some reagents Need clean glassware	• More efficient thus modern equipment have automatic data handling hence more reliable Advantages Automated equipment makes data/statistical handling easy Require small volume of solution Disadvantages Expensive equipment Complex procedures Require skillful operator
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Analytical methods based on sample size

Method	Minimum size	Gram
Macro	100mg	10^{-1}
Semi macro	10mg	10^{-2}
Micro	1mg	10^{-3} miligram
Ultra micro	1ug	10^{-6} microgram
Sub microgram	10^{-2} ug	10^{-8}
		10^{-9} nanogram
		10^{-12} picogram
		10^{-15} femtogram
		10^{-18} attogram

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Problems of micro/trace analysis

- The **interferences**:-other compounds present in the sample and can be detected by the method used giving erroneous results. These should be removed or eliminated before determination.
- Contamination**:-presence of other analytes from elsewhere during analysis such as from storage containers, working apparatus and reagents. Can be minimized by:-
 - Use of specific containers for specific compounds such as polythene bottles for mineral ions or glass bottles for organic compounds are used.
 - Carrying out **blank** analysis. A blank is a solution or extract that contains all other reagents and use similar apparatus but does not contain the analyte (the substance being analysed). $[Cu] = [sample] - [blank]$.
 - Reagents are supplied with specific standards that are meant to reduce or estimate contamination.

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- c) Sample **loses** eg by sticking on the wall of containers, especially organic materials where upto 70% can be lost on the surface of the container.

Criterion for selection of methods of Analysis

speed (the time required to carry out the analysis), availability of equipment in the field or laboratory, sample size available, concentration of material being analyzed, nature of sample (its solubility, toxicity, radioactivity, value), accuracy and precision of data required, cost of analysis and identity of sample (whether to use destructive or non-destructive method).

Ways of expressing the amount or the concentration of analyte in the sample

Results are reported as concentration on either weight or volume basis, that is, the quantity of analyte per unit weight or per unit volume. The units used are;

- **Weight:** gram (g), Kilogram ($\text{kg} = 10^3\text{g}$), milligram ($\text{mg} = 10^{-3}\text{g}$), microgram ($\mu\text{g} = 10^{-6}\text{g}$), nanogram ($\text{ng} = 10^{-9}\text{g}$)
- **Volume:** litre(l), millilitre ($\text{ml} = 10^{-3}\text{l}$), microliter ($\mu\text{l} = 10^{-6}\text{l} = 10^{-3}\text{ml}$), nanoliter ($\text{nl} = 10^{-9}\text{l} = 10^{-6}\text{ml}$)
- **Concentration:** Concentration include parts per thousand (ppt), parts per million (ppm), parts per billion (ppb), molarity (M), Normality (N).

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Calculation and interpretation of analytical data

Analytical procedures are evaluated through the calculation and interpretation of analytical data and is important because it is a measure of performance and confidence of the analytical procedure. Good **laboratory records** are essential in analytical chemistry. This ensures that the correct date is used for calculation. This is important when communicating the validity of the results.

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Errors in Chemical Analysis

Results must be accompanied with some estimate of associated errors. An **error** is the amount of variation found or expected in a measurement under a given experimental condition.

$\text{Error} = \text{Observed value} - \text{True value}$

Classification

Random (indeterminate) errors:-

- source cannot be positively identified or magnitude estimated.
- cause individual results to fall on both sides of the expected measurement normal/Gaussian distribution curve.
- revealed by a small difference in successive measurements made by the same analyst under identical conditions..
- difficult for the analyst to control or make corrections or even to recognize factors that will result in the errors.
- The source is part of all scientific measurements due to the inability to control all variables.
- Random errors affect the precision or reproducibility and repeatability of an experiment.

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Systematic (determinate) errors

- all the results are in error in the same sense (either high or low) that is, unidirectional.
- can be avoided or corrected once the source is identified.
- They are related to improper design or adjustment of experimental conditions
- affect accuracy of a result.

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Home Exercise: To distinguish random and systematic errors. Students W, X, Y and Z performed five replicate titrations on 10.00ml of 0.1M NaOH against 0.1HCL the results were as follows;

Student	Results	Mean/range	Comment on errors/accuracy/precision
W	10.08, 10.11, 10.09, 10.10, 10.12	Mean=10.10 Sd= 0.016 Range= 0.04	
X	9.88, 10.14, 10.02, 9.80, 10.21	Mean= 10.01 Sd= 0.172 Range=0.41	
Y	10.19, 9.79, 9.69, 10.05, 9.78	Mean= 9.90 Sd= 0.210 Range= 0.05	
Z	10.04, 9.98, 10.02, 9.97, 10.04	Mean= 10.01 Sd= 0.033 Range= 0.07	

NOTE

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- Random and systematic errors can occur independently of one another and may thus arise at different stages of the experiment.
- Random errors cannot be eliminated though by careful technique can be minimized and by making repeated measurements they can be measured and their significance evaluated.
- Systematic errors may be removed by careful checks of the experimental technique and equipment

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Sources of systematic errors

- a) Instrumental sources such as faulty or uncalibrated equipment
- b) Operative (personal) sources are from the analyst or on the procedure such as colour blindness, wrong way of handling instrument, wrong transfer of solution, incomplete drying of samples or a mathematical errors in calculating weights or volumes used. These can be reduced by experience and care by the analyst during an experiment.
- c) Methodic sources involve the chemical properties of the system such as solubility of a precipitate, side reaction, incomplete reactions, impurities in reagents. These cannot be

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changed unless the conditions of the determination are altered but may be corrected by running a blank analysis.

Precautions to minimize errors

- Carefully consideration of each step of the experiment and the apparatus to be used so as to calibrate them where necessary, and the sampling and analytical procedures so as to identify the likely source of systematic errors.
- Design the experiment at every stage to reduce systematic errors, such as weighing by difference can the use of calibration curve on absorption measurements to reduce absorbance errors and path-length inaccuracies.
- Use of standard reference materials and methods

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The scientific calculator

- Deleting memory before entry of any data (remove M+)
- Setting to statistical mode
- Entering data
 - Data 1:- 1.80, 1.58, 1.64, 1.77, 1.58
 - Data 2:- 0.084, 0.089, 0.079, 0.080
- Retrieving the mean
 - Data 1 _____
 - Data 2 _____
- Retrieving the standard deviation for small population
 - Data 1 _____
 - Data 2 _____
- Checking for n
- Deleting data for new entry

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Precision and Accuracy

Analysis of a sample is generally repeated between two to five times to get **replicate** measurements.

Statistical terms

- Mode** is the most frequently observed value or values in an analysis. When it is one values the distribution is monomodal, indicating homogeneity. A bimodal distribution of the experimental values indicate that the sample is heterogeneous.
- Mean** is sometimes referred to as the arithmetic mean or average is symbolized as $(\bar{x})$. it is the value obtained when replicated values are added and divided by the number of

measurements (n) in a set of data.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

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c) **Median** is the value about which all other values are equally distributed so that half of the values are larger and half are smaller. For an odd set of data the middle value of data when it is arranged in ascending or descending order gives equal number of values measured on both sides of one value which is thus the median. For an even number of data set the median is the average of the middle pair when data is arranged in ascending or descending order.

d) **Average deviation** ($\bar{d}$) is found by summing the individual deviations, and dividing by the number of measurements. Individual deviations are simply the difference, taken without regards to sign, between the experimental values and mean. It is a measure of the scatter of

measurements.

$$\bar{d} = \frac{\sum |x_i - \bar{x}|}{n}$$

e) **Standard deviation** is the root mean square deviation. It is the most useful measure of the spread of analytical data about the mean and is a **better** measure of precision than the average deviation because it gives more weight to large deviations.

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

for small measurements (data) with (n-1) being a bias correcting term known as 'degree of freedom'.

f) **Average deviation of the mean** is the average deviation divided by the total number of measurements.

$$\bar{d}(\text{mean}) = \frac{\bar{d}}{n}$$

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g) **Standard deviation of the mean/ standard error of the mean**, is the standard deviation divided by the root of the total number of measurement.

$$s(\text{mean}) = \frac{s}{\sqrt{n}}$$

This implies that the precision of the mean of a series of n measurements is inversely proportional to the square root of n of the individual values.

Note

a) Both average deviation and standard deviation can be expressed in relative terms to facilitate comparison between data sets thus

b) **Relative average deviation** = $\frac{\bar{d}}{\bar{x}} \times 100$

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c) **Relative standard deviation (RSD)** also known as Coefficient of Variation (CV) is the standard deviation expressed as a percentage of the mean. It is used in the comparison of the precisions of results that have different units or magnitudes. The units therefore are in percent form.

$$RSD/(CV) = \frac{s}{\bar{x}} \times 100$$

d) Relative values can be expressed fractionally as parts per hundred or percent (multiplied by 100), parts per thousand (multiplied by 1000), etc

e) The values for mean, standard deviation, number of measurements, and sum of measurements can be obtained using a scientific calculator or a spread sheet.

f) Range (R)

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Example :-The following weights were obtained in chemical determination of calcium in a sewage effluent sample: - 29.8, 30.2, 28.6, and 29.9mg. Calculate

- The average deviation and standard deviation of the individual values (express these as absolute and relative values).
- The average deviation and standard deviation of the mean

Solution

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{118.3}{4} = 29.6mg$$

$$\bar{d} = \frac{\sum |x_i - \bar{x}|}{n} = \frac{1.9}{4} = 0.485mg$$

$$\bar{d}(\text{relative}) = \frac{\bar{d}}{\bar{x}} \times 100 = \frac{0.485}{29.6} \times 100 = 1.6\%$$

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$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{1.41}{4-1}} = 0.69$$

$$RSD / CV = \frac{s}{\bar{x}} \times 100 = \frac{0.69}{29.6} \times 100 = 2.33\%$$

$$\bar{d}(\text{mean}) = \frac{\bar{d}}{n} = \frac{0.48}{4} = 0.12$$

$$s(\text{mean}) = \frac{s}{\sqrt{n}} = \frac{0.69}{\sqrt{4}} = 0.35$$

$$\text{Range} = (\text{highest} - \text{lowest}) = (30.2 - 28.6) = 1.6$$

Precision:- the degree of agreement between replicate measurements in a set of data that have been made in an identical fashion. The amount of scatter or variation of data. It measures the repeatability or reproducibility of determinations. Precision can be expressed or measured in terms of deviation from the mean, deviation from the median, standard deviation and spread or range.

Repeatability is when measurements are done using same apparatus and at the same time.

Reproducibility is when measurements are done using same apparatus but at different times.

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Accuracy:-The degree of agreement between the measured value (experimental value), denoted x_i and the true value (accepted value), denote by μ . The departure of a determination from the true value. The true or accepted true value is the known or given. Its expressed as in terms of absolute error or relative error.

a) **Absolute error** is the difference between the measured value x_i and the true value with regard to sign. Absolute error = $x_i - \mu$

b) **Mean error** (average error) individual absolute errors are used. $Error(mean) = \frac{\sum(x_i - \mu)}{n}$

c) **Relative error:** is the absolute error or mean error expressed as a ration of the true value.

$$Error(relative) = \frac{x_i - \mu}{\mu} \times 100 \quad \text{or} \quad Error(relative) = \frac{error(mean)}{\mu} \times 100$$

NOTE

- The relative error can be expressed in percentage form. Thus the percent error is the relative error multiplied by 100.
- If the relative error is multiplied by 100, this is called parts per hundred error (pph), multiplied by 1000, this is called parts per thousand error (ppt), by one million called parts per million (ppm) and by a billion, parts per billion (ppb) etc.
- The precision of a measurement can be improved by increasing the number of observations. Good precision doesn't assure good accuracy. This occurs when there is a systematic error in the analysis
- The higher the degree of precision, the greater the chance of obtaining the true value. That is the more agreed the replicate measurements are the higher the chance of getting the accepted true value
- The more measurements that are made, the more the reliable will be the measure of the precision.

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Example :-The results of an analysis are 36.97% compared with the accepted value of 37.06%. What is the relative error in parts per thousand (ppt).

$$Error(absolute) = (36.97 - 37.06) = -0.09\%$$

$$Error(relative) = \frac{-0.09}{37.06} \times 1000 = -2.4ppt$$

Rejection of Outlying Values

When a series of replicate analysis (measurements) is performed, one of the measurements may appear to differ from others in an extraordinary way (an outlier). Statistical tests to enable make decision on rejection of a data entry:-

The Q- Test

(i) Arrange the data in ascending or descending order _____

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- (ii) Calculate a value Q as follows:- $Q_{cal} = \frac{|suspect - nearest|}{range\ of\ data} = \underline{\hspace{2cm}}$
- (iii) Read the **Q-table** for n measurements at 95%CL unless stated otherwise = $\underline{\hspace{2cm}}$
- (iv) Compare the calculate value and tabulated value by inserting the right symbol $Q_{cal} \underline{\hspace{0.5cm}} Q_{tab}$
- (v) Conclude based on whichever applies between:-
- i) $Q_{cal} \geq Q_{tab} \Rightarrow reject\ value(outlier)$
- ii) $Q_{cal} < Q_{tab} \Rightarrow retain\ value(not\ outlier)$
- Conclusion** $\underline{\hspace{3cm}}$

Example :--Use the 90% confidence level to determine if the last value of the following data for % Fe in a sample should be omitted. 2.93, 3.08, 3.11, 3.04, 2.70%

Solution

Arranging data 2.70 2.93 3.04 3.08 3.11

$$Q_{cal} = \frac{|suspect - nearest|}{range} = \frac{|2.70 - 2.93|}{3.11 - 2.70} = \frac{0.23}{0.41} = 0.56$$

$$Q_{tab}(90\%) = 0.64(n = 5)$$

$$Q_{cal} < Q_{tab}$$

Therefore, the value is retained

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3s-steps

- (i) Calculate mean and standard deviation without the suspect value: $\bar{x} = \underline{\hspace{1cm}}$ $s = \underline{\hspace{1cm}}$
- (ii) Calculate 3 times the standard deviation to obtain 3s value = $\underline{\hspace{2cm}}$
- (iii) Calculate $|suspect - \bar{x}|$ and 3s by inserting the correct symbol $|suspect - \bar{x}| \underline{\hspace{0.5cm}} 3s$
- (iv) Conclude based on whichever applies between:-
- i) $|suspect - \bar{x}| \geq 3s \Rightarrow reject\ value(outlier)$
- ii) $|suspect - \bar{x}| < 3s \Rightarrow retain\ value(not\ outlier)$
- Conclusion** $\underline{\hspace{3cm}}$

Example:- The following set of chloride ion concentration (mmol) analysed on separate aliquots of a pooled serum reported 103,106,107,114. Determine whether the value 114 should be rejected using 3s test.

Solution

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$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = 105.3$$

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = 2.08$$

$$3s = 3 \times 2.08 = 6.24$$

$$|\text{suspect} - \text{mean}| = 114 - 105.3 = 8.7$$

$$8.7 > 6.24$$

The suspect value is rejected

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4 $\bar{d}$ -test steps

- (i) Calculate the mean and average deviation ($\bar{d} = \frac{\sum |x_i - \bar{x}|}{n}$) without the suspect value:-

$$\bar{x} = \underline{\hspace{2cm}} \quad \bar{d} = \underline{\hspace{2cm}}$$

- (ii) Calculate 4 times the standard deviation to obtain 4 $\bar{d}$ value = $\underline{\hspace{2cm}}$

- (iii) Calculate $|\text{suspect} - \bar{x}|$ and 4 $\bar{d}$ by inserting the correct symbol $|\text{suspect} - \bar{x}| \underline{\hspace{1cm}} 4\bar{d}$

- (iv) Conclude based on whichever applies between:-

$$i) |\text{suspect} - \bar{x}| \geq 4\bar{d} \Rightarrow \text{reject value(outlier)}$$

$$ii) |\text{suspect} - \bar{x}| < 4\bar{d} \Rightarrow \text{retain value(not outlier)}$$

Conclusion $\underline{\hspace{4cm}}$

Example:- The following was measurement of chloride concentration 29.8, 30.2, 28.6, 29.7mg/l

Use the 4d test to determine whether 28.6 is suspect.

Solution

$$\bar{x} = \frac{\sum x_i}{n} = 29.9 \text{mg / l}$$

$$\bar{d} = \frac{\sum |x_i - \bar{x}|}{n} = 0.2$$

$$4\bar{d} = 4 \times 0.2 = 0.8$$

$$|28.6 - 29.9| = 1.3$$

$$1.3 > 4\bar{d}$$

Therefore the value is rejected.

Note:- All the three rejection tests may give different conclusions on a suspect value

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Confidence Limits of the mean

These are the lower and upper values for which the true value is found within a probability level(%). It is obtained by calculating the confidence interval (the range within which the true value lies)

$$\text{Confidential interval } \mu = \bar{x} \pm \frac{ts}{\sqrt{n}}$$

$$\text{Confidential limits } \bar{x} - \frac{ts}{\sqrt{n}} \text{ and } \bar{x} + \frac{ts}{\sqrt{n}}$$

Example:-What are the 95% confidence interval and confidence limits for the data 102, 97, 99, 98, 101, 106.

Solution

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Using the calculator, mean = 100.5 and s = 3.27. From the t-table at (n-1), t=2.57

$$\mu = \bar{x} \pm \frac{ts}{\sqrt{n}}$$

$$\mu = 100.5 \pm \frac{2.57(3.27)}{\sqrt{6}}$$

$$\mu = 100.5 \pm 3.4$$

$$\bar{x} - \frac{ts}{\sqrt{n}} \text{ and } \bar{x} + \frac{ts}{\sqrt{n}}$$

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97.1 and 103.9

Tests of significance

Much of scientific work is based on hypothesis testing. If the results do not support the hypothesis, it is rejected and may require a new hypothesis while if supported it serves as a basis for further experiments. Scientists therefore judge whether a numerical difference is a manifestation of random errors or not. Some statistical tests employed such as the **student t-test** and **F-tests** make use of the **null hypothesis** stated “there is no significant difference between any two sets of measurements’. The probability of the observed differences is computed to justify the hypothesis or not usually at 95% confidence level.

a) The student t-test

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This is employed in checking for **accuracy** in the following cases;

- (i) To compare the mean of an analysis with a known/true value. Example _____
- (ii) Comparing two means from analysis of the same sample analysed by two methods/analyst etc. Example _____
- (iii) Comparing two means from analysis of individual samples analysed by two methods/analyst etc (Paired t-test). Example _____

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Note

- The null hypothesis is tested and conclusion made such that in the above cases, the analyst is accurate (there are no systematic errors) if the hypothesis is justified for (i) and that the two sets of data obtained do not differ in their means if the hypothesis is justified for (ii) and (iii).
- t values are read for various degrees of freedom and at 95% confidence level unless otherwise stated. **(Reading of t table).**

i) Comparing an experimental mean with a true value

For a standard reference material containing 38.9% Hg, the values obtained are 38.9, 37.4, and 37.1%. Determine whether systematic errors were involved.

Steps

- Obtain the mean = _____ and s = _____
- Calculate t using equation: $|t_{cal}| = (\bar{x} - \mu) \frac{\sqrt{n}}{s} = \underline{\hspace{2cm}}$
- Read the t value for (n-1) degrees of freedom at 95% to obtain the tabulated value t_{tab} :-
= _____
- Compare the calculated and tabulated value by inserting the right symbol $|t_{cal}| \underline{\hspace{1cm}} t_{tab}$
- Conclude based on whichever applies between:-
 - $t_{cal} \geq t_{tab} \Rightarrow \text{reject hypothesis (unjustified)}$
 - $t_{cal} < t_{tab} \Rightarrow \text{retain hypothesis (justified)}$

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Conclusion _____

Q. Is the analyst accurate? _____

ii) Comparing two experimental means

The null hypothesis is tested to check whether there is real discrepancy in the numerical values or due to random errors.

Q. The composition of a flake of paint found on a victims clothes after a hit and run accident was compared with that of paint from a suspected car. The results of analysis for titanium from the paint on the clothes and from the car were as follows:-

Data 1:- Paint from clothes 4.0, 4.6, 4.3, 4.2

Data 2:- Paint from car 4.5, 5.3, 5.5, 5.0, 4.9

Do the analysis suggest a difference in composition? (is the sample the same? Is the hypothesis justified? Did the suspect car hit the victim? Do the means differ?)

Steps SIR OTONDE MICHAEL AWITI

- Calculate the means, standard deviations and variances (s^2) from both analysis

$$\text{Data 1: } \bar{x}_1 = \underline{\hspace{2cm}} s_1 = \underline{\hspace{2cm}} s_1^2 = \underline{\hspace{2cm}}$$

$$\text{Data 2: } \bar{x}_2 = \underline{\hspace{2cm}} s_2 = \underline{\hspace{2cm}} s_2^2 = \underline{\hspace{2cm}}$$

Calculate a pooled standard deviation using equation:

-

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$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n - 1)s_2^2}{n_1 + n_2 - 2}} = \underline{\hspace{2cm}}$$

$$|t|_{cal} = \frac{\bar{x}_1 - \bar{x}_2}{s_p} \sqrt{\frac{n_1 n_2}{n_1 + n_2}} = \underline{\hspace{2cm}}$$

- (iii) Calculate t value using equation $t_{cal} = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{n_1 + n_2}}$
- (iv) Read the critical t value for $(n_1 + n_2 - 2 \text{ degrees of freedom}) = \underline{\hspace{2cm}}$
- (v) Compare the calculated and tabulated value by inserting the right symbol
 $|t_{cal}| \underline{\hspace{1cm}} t_{tab}$

Conclude based on whichever applies between:-

$$(y_i)_{cal} \geq t_{tab} \Rightarrow \text{reject hypothesis (unjustified)}$$

$$ii) t_{cal} < t_{tab} \Rightarrow \text{retain hypothesis (justified)}$$

Conclusion

Q. what assumption is behind pooling of data?

Q. What is the advantage of pooling data?

Example:-Test whether the mean results obtained by these two methods differ significantly.

	New method	official method
Mean	19.65%	19.24%
Standard deviation	0.673%	0.324%
Sample size	6	5

Solution SIR OTONDE MICHAEL AWITI

The means and standard deviation are obtained from the calculator.

$$s_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

$$S_p = \sqrt{\frac{(6-1)(0.673)^2 + (5-1)(0.324)^2}{6+5-2}} = 0.546$$

$$|t_{cal}| = \frac{\bar{x} - \bar{y}}{S_n} \sqrt{\frac{n_1 n_2}{n_1 + n_2}} = \frac{19.65 - 19.24}{0.546} \sqrt{\frac{6 \times 5}{6 + 5}} = 1.239$$

$$t_{tab} = 2.262$$

Since $t_{cal} < t_{tab}$, the means do not differ significantly.

iii) Paired t-test

Q. The concentration of cadmium in four samples were determined using two methods with results as tabulated-Do the means from the two methods differ?

Steps

- (i) Obtain (with respect to sign) the individual differences by subtracting the later data from the former to obtain a new data set (d_i)

Sample	Wet digestion	Direct digestion	Individual difference= d_i
1	71	76	
2	68	61	
3	48	50	
4	60	57	

- (ii) Calculate the mean and standard deviation for the new data set:

$$\bar{x} = \text{_____} \quad s = \text{_____}$$

- (iii) Calculate t using the equation: $|t|_{cal} = \frac{\bar{x}}{s} \sqrt{n} = \text{_____}$

- (iv)

Read the critical value of t for (n-1) degrees of freedom= _____

- (v) Compare the calculated and tabulated value by inserting the right symbol

$$|t_{cal}| \text{ _____ } t_{tab}$$

- (vi) Conclude based on whichever applies between:-

$$i) t_{cal} \geq t_{tab} \Rightarrow \text{reject hypothesis (unjustified)}$$

$$ii) t_{cal} < t_{tab} \Rightarrow \text{retain hypothesis (justified)}$$

Conclusion _____

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b) Comparison of precision (the F-test)

This is used to test the null hypothesis by comparing two data sets in terms of their variances/precisions.

Q. What is the relationship between precision and standard deviation?

Example:- Test whether the standard deviations obtained by these two methods differ significantly.

	New method	official method
Mean	19.65%	19.24%
Standard deviation	0.673%	0.324%
Sample size	6	5

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Steps

- Obtain the variances for each method: $s^2 = \text{_____}$ and $s^2 = \text{_____}$
- Label subscripts for the variances with subscript 1 for that for higher variance:
 $s_1^2 = \text{_____}$ and $s_2^2 = \text{_____}$
- Calculate F value (always >1) using equation: $F_{cal} = \frac{S_1^2}{S_2^2} = \text{_____}$
- Read** the F tabulated value for $\nu_1 = \text{_____}$ $\nu_2 = \text{_____}$ $F_{tab} = \text{_____}$

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- v. Compare the calculated and tabulated value by inserting the right symbol
 $F_{cal} \text{ ______ } F_{tab}$
- vi. Conclude based on whichever applies between:-
i) $F_{cal} \geq F_{tab} \Rightarrow \text{reject hypothesis(unjustified)}$
ii) $F_{cal} < F_{tab} \Rightarrow \text{retain hypotheis(justified)}$

Conclusion _____

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